Solution Requirements (Functional & Non-functional)

Date	31 January 2026
Team ID	Not provided — confirm with MySkillWallet/college
Project Name	EV Market Intelligence Dashboard
Maximum Marks	4 Marks

Functional Requirements

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)	Priority (High/Medium/Low)
1	Data Access & Integration	Connect to the supplied EV market CSV dataset containing category, manufacturer, manufacturing country, battery type, model year, annual sales, battery capacity, range, customer rating, and CO■ savings fields.	High
2	Data Processing	Verify field types; use dimensions for category/manufacturer/country/battery type/model name/model year and measures for sales, capacity, range, rating, and CO■ savings. Use appropriate aggregation and handle missing or invalid values transparently.	High
3	Business Analysis	Aggregate annual sales by category and manufacturer; identify top 10 manufacturers; compare battery technology distribution, range by category, average customer rating, sales by model year, capacity-versus-range relationship, and CO■ savings by category.	High
4	Business Insights & Analytical Outputs	Provide nine views: (1) sales by category, (2) top 10 manufacturers by sales, (3) battery technology mix, (4) driving range by category, (5) manufacturing-country map, (6) customer satisfaction by category, (7) sales by model year, (8) battery capacity vs. range scatter plot, and (9) sustainability impact by category.	High
5	User Interaction & Analysis	Allow users to sort and inspect bars, use available filters/highlighting, compare categories and manufacturers, and view details through tooltips. Interactions should remain understandable in the published dashboard.	Medium
6	Data Update & Reporting	Publish the dashboard to Tableau Public and include its URL in the project documentation. Save the Tableau workbook and export the completed project templates/documentation as Word and PDF. Refresh by replacing/reloading the source dataset when an updated file is provided.	Medium

Non-functional Requirements

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Use clear chart titles, readable labels, consistent units, and a layout that allows the nine visualizations to be understood without excessive scrolling. Provide useful tooltips and avoid overlapping views.
NFR-2	Security	Use the provided dataset only. Do not include personal or confidential data. Tableau Public workbooks and published dashboards are public, so only publish material approved for public viewing.
NFR-3	Reliability	Ensure the workbook opens after publication, worksheets reference valid fields, chart aggregations are consistent, and the published dashboard link works. Verify all nine views before submission.
NFR-4	Performance	Avoid unnecessary marks and overly complex views. Use a sensible dashboard size, limit the manufacturer chart to the top 10, and filter or aggregate data when the browser becomes slow.
NFR-5	Availability	The dashboard is accessible through Tableau Public when the service and public link are available. The project also retains a local workbook copy for review.
NFR-6	Scalability	The current solution is a student-scale dashboard using a CSV file. Larger or frequently refreshed datasets may require data extracts, stronger aggregation, or a managed database; those capabilities are outside the current scope.